

Nanotechnology

Engineering matter at the billionth-of-a-metre scale

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Nanotechnology is the science of designing, building, and using materials and devices at the nanoscale — roughly 1 to 100 nanometres, where a single nanometre is about 100,000 times thinner than a human hair. At this size, familiar materials behave differently, opening possibilities that don't exist at everyday scales.

DEFINITION

The manipulation of matter at atomic, molecular, and supramolecular scales to create structures, devices, and systems with novel properties and functions.

WHY IT MATTERS

Nanoscale materials often have dramatically different mechanical, optical, electrical, or chemical properties than the same material in bulk form — enabling breakthroughs across medicine, electronics, energy, and manufacturing that aren't achievable any other way.

WORKING PRINCIPLE



As particle size shrinks toward the nanoscale, a much larger fraction of atoms sit at the surface rather than the interior. This surface dominance, combined with quantum mechanical effects, changes how a material conducts electricity, absorbs light, and reacts chemically — properties engineers deliberately exploit.

QUICK FACTS

- ▶ A sheet of paper is about 100,000 nanometres thick
- ▶ The term was popularized after physicist Richard Feynman's 1959 talk 'There's Plenty of Room at the Bottom'
- ▶ Gold nanoparticles can appear red or purple, not gold, due to quantum effects

KEY TERMS

- Nanoparticle
- Quantum dot
- Self-assembly
- Nanocomposite
- Top-down fabrication
- Bottom-up fabrication

CORE CONCEPTS

- ▶ Quantum confinement effects
- ▶ Surface-area-to-volume ratio
- ▶ Self-assembly of nanostructures
- ▶ Top-down vs. bottom-up fabrication

APPLICATIONS

- ▶ Targeted drug delivery capsules
- ▶ Scratch-resistant and water-repellent coatings
- ▶ High-capacity battery electrodes
- ▶ Water filtration membranes
- ▶ Faster, smaller transistors

REAL-LIFE EXAMPLES

- ▶ Silver nanoparticles in antibacterial wound dressings
- ▶ Titanium dioxide nanoparticles in sunscreen
- ▶ Carbon nanotubes reinforcing bicycle frames

ADVANTAGES

- ▶ Enables materials with tailored properties
- ▶ Reduces material usage for the same performance
- ▶ Opens entirely new device categories

LIMITATIONS

- ▶ High production costs at scale
- ▶ Potential health and environmental exposure risks not fully understood
- ▶ Difficult to manufacture with atomic precision reliably

CAREER OPPORTUNITIES

Nanomaterials research scientist, Semiconductor process engineer, Nanomedicine formulation specialist, Materials characterization technician

INDUSTRY APPLICATIONS

Electronics & semiconductors, Pharmaceuticals, Energy storage, Textiles & coatings

RESEARCH AREAS

Nanotoxicology, Nanophotonics, DNA nanotechnology, Nanocatalysis

FUTURE SCOPE

Growing convergence with AI-driven materials discovery and biotechnology is expected to accelerate development of programmable nanomaterials for medicine and sustainable energy over the next decade.

CURRENT TRENDS

Increasing focus on green nanotechnology synthesis methods and AI-assisted design of nanostructures with predictable, safer properties.

SUMMARY

Nanotechnology harnesses the unusual behavior of matter at the nanoscale to create materials and devices with properties impossible at ordinary size — already reshaping medicine, electronics, and energy, with much more still ahead.